Alex Rivera

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Projects

Synchronous Buck Converter for a CubeSat Power Board Fall 25 - Spring 26

KiCad, LTspice, STM32, Embedded C, bench testing with an electronic load

- Reached 92% peak efficiency at a 2 A load, measured on an electronic load, by replacing the catch diode with a synchronous MOSFET stage and tuning dead time to 40 ns with an STM32 timer, up from 84% before
- Cut output ripple from 85 mV to 18 mV peak to peak by redesigning the 4-layer PCB in KiCad with a solid ground plane and a tighter switching loop, then resizing the LC output filter for 500 kHz switching
- Wrote STM32 firmware in C that samples input current at 1 kHz and shuts the rail down within 50 µs of an overcurrent fault, which kept the board alive through 3 deliberate short-circuit tests at a full 12 V input
- Kept the FETs under 50 °C at full load by adding thermal vias and a 2 oz copper pour under them

FPGA Five-Band Audio Equalizer Spring 25

Verilog, Vivado, MATLAB, Basys 3 FPGA, fixed-point DSP

- Processed 48 kHz stereo audio at 0.4 ms latency by pipelining five 64-tap FIR filters in Verilog
- Matched the MATLAB response within 0.1 dB from 20 Hz to 20 kHz by testing coefficients in Vivado
- Cut LUT use 35% by sharing one multiplier per filter across all 64 taps with time-division scheduling

Bluetooth Battery Monitor Fall 24

ESP32, Arduino C++, Python, INA219 current sensor

- Tracked 4 Li-ion cells within 10 mV each with an ESP32 monitor that streams to a Python dashboard
- Extended pack runtime 15% in field tests by adding a cutoff that drops non-critical loads at 3.3 V

Experience

Power Subsystem Engineer, State University CubeSat Team Fall 24 - Spring 26

- Cut power board mass by 30 g by replacing 6 linear regulators with one buck converter board, which then passed thermal vacuum testing from -20 °C to 60 °C on the first attempt with no board rework needed
- Found 2 battery protection faults before launch review by writing a Python test script that runs 200 charge and discharge cycles overnight and flags any cell that drifts more than 20 mV from the pack average

Undergraduate Research Assistant, Power Electronics Lab, State University Summer 25

- Reduced GaN switching loss measurement error from 12% to 3% by adding probe deskew calibration to the double-pulse test setup, which the lab now uses as the standard setup for every GaN device it tests
- Cut test time per device from 40 to 8 minutes by automating oscilloscope capture with Python and PyVISA

Education

B.S. Electrical Engineering, GPA: 3.62/4.00 Expected Spring 27

State University

Relevant coursework: Signals and Systems, Electronics I and II, Power Electronics, Digital Design, Embedded Systems, Control Systems, Electromagnetics, Probability for Engineers, Microcontroller Lab

Activities: Tau Beta Pi, IEEE student branch

Skills

Certifications: CSWA (Certified SolidWorks Associate), IPC J-STD-001 soldering

Software: KiCad, LTspice, MATLAB, Vivado, Python, C, Verilog, SolidWorks, Git